Can Alkan

CONTACT INFORMATION

Department of Computer Engineering
Bilkent University
Engineering Building, EA 509
Bilkent, Ankara 06800
Turkey

E-mail: calkan@cs.bilkent.edu.tr
Web: http://www.alkanlab.org
ORCID: 0000-0002-5443-0706
ResearcherID: D-2982-2009
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RESEARCH INTERESTS

Combinatorial algorithms for analyzing biological sequences, genomic structural variation, pangenomics, and hardware acceleration of bioinformatics workloads using processing-in-memory, NPU, GPU, and FPGA.

EXPERIENCE

Department of Computer Engineering, Bilkent University, Bilkent, Ankara, Turkey

Professor **September 2026 - present**
Associate Professor **April 2021 - June 2026**
Assistant Professor **January 2012 - April 2021**

Seven Bridges Genomics, Cambridge, MA, United States

Visiting Scholar (on sabbatical) **September 2018 - February 2019**

Department of Computer Science, ETH Zürich, Zürich, Switzerland

Visiting Professor (on sabbatical) **February - July 2018**

Department of Genome Sciences, University of Washington, Seattle, Washington, USA

Acting Assistant Professor **June 2011 - January 2012**
Senior Fellow **October 2005 - May 2011**

School of Computing Science, Simon Fraser University, Burnaby, British Columbia, Canada

Visiting Researcher **January, 2004 - September 2005**

EDUCATION

University of Washington, Seattle, Washington, USA
Department of Genome Sciences and Howard Hughes Medical Institute

Postdoctoral Fellow, Genomics, October 2005 - December 2011

- Worked on computational methods to reconstruct the evolutionary history of alpha-satellite DNA, *de novo* alphoid sequence prediction from whole-genome shotgun sequence data and detection of human structural variation and segmental duplications using next-generation sequencing technologies.
- Advisor: Evan E. Eichler

Case Western Reserve University, Cleveland, Ohio, USA
Department of EECS

Ph.D., Computer Science, August 2005

- Dissertation Topic: "Computational Studies on Evolution and Functionality of Genomic Repeats"
- Advisor: S. Cenk Şahinalp

Bilkent University, Ankara, Turkey
Department of Computer Engineering

B.Sc., Computer Engineering and Information Science, May, 2000

Honors and Awards

- TÜBİTAK Incentive Award, 2015.
- Thomson Reuters Highly Cited Researcher, 2014, 2018.
- Young Investigator Award (BAGEP), Science Academy of Turkey, 2013.
- AAAS Newcomb Cleveland Prize for the Most Outstanding Paper Published in Science, American Association for the Advancement of Science, USA, 2010.

Professional Memberships

- International Society for Computational Biology

Research Grants

Active

- Turkish Institutes of Health (TÜSEB), 2026-2028
  Title: Rapid pathogen detection with mobile computing platforms.
  Goal: Sequencing-based identification of pathogens quickly using NPU-enabled laptops and Jetson Nano kits.
  PI: Can Alkan

Previous

- Horizon Europe EIC Pathfinder, 2022-2026
  Title: Processing-in-memory architectures and programming libraries for bioinformatics algorithms (BioPIM).
  Goal: Accelerate bioinformatics applications through leveraging the Processing-in-Memory (PIM) technologies.
  Coordinator: Can Alkan
  Team Leaders: Onur Mutlu (ETH Zürich), Rayan Chikhi (Institut Pasteur), Dominique Lavenier (CNRS), Abu Sebastian (IBM Zürich), Ran Ginosar (Technion), Yann Falevoz (UPMEM), Leonid Yavits (Bar-Ilan).
- NVIDIA Applied Research Accelerator Program, 2023-2025
  Title: Fast pathogen screening using DNA sequencing in disaster-stricken areas.
  Goal: Sequencing-based identification of pathogens quickly using Jetson Nano kits.
  PI: Can Alkan
- Afyon Health Sciences University, 2020-2021
  Title: Developing rapid diagnostic kit for sepsis (Sepsi-Fast).
  Goal: Determine antibiotic resistance and susceptibility in newborn sepsis cases.
  PI: Nurullah Okumuş
- Scientific and Technical Research Council of Turkey (TÜBİTAK-2546-115E596), 2017-2019
  Title: BacGenTrack – an integrated system for bacterial genome tracking using high-throughput sequencing technology: from identification to visualization.
  Goal: Develop a user-friendly web system and novel analysis algorithms needed to facilitate the analysis and provide the essential data sharing tools necessary for its effective use in the field of molecular epidemiology.
  PI: Can Alkan and Joao Andre Carriço
- European Molecular Biology Organization Installation Grant (IG-2521), 2013-2018
  Title: Development and application of computational methods to analyze next-generation sequence data to characterize both normal and disease-causing variation and build *de novo* genome assemblies.
  Goal: Analysis of biological sequences generated with the next-gen sequencing platforms.
  PI: Can Alkan
- Scientific and Technical Research Council of Turkey (TÜBİTAK-1001-215E172), 2016-2018
  Title: Algorithms for structural variation discovery using hybrid sequencing technologies and library preparation protocols.
  Goal: Discovery and characterization of structural variants using multiple sequencing platforms, linked-reads, and read clouds.
  PI: Can Alkan

- Scientific and Technical Research Council of Turkey (TÜBİTAK-2507-214Z130), 2015-2018
  Elucidating the mechanisms of sequential trastuzumab/T-DM1 resistance *in vitro* and *in vivo* models of HER-2 overexpressing breast cancer.
  Goal: Unraveling molecular mechanisms of acquired T-DM1 resistance in trastuzumab-resistant breast cancer models and to regain T-DM1 sensitivity. I served as a researcher for the bioinformatics components of this project.
  PI: Özgür Şahin
- Scientific and Technical Research Council of Turkey (TÜBİTAK-1001-115O391), 2015-2018
  Title: *de novo* genome sequencing and identification of genes involved in biotic-abiotic stress of wild beet species (Beta Corolliflora and Beta Procumbens), and cloning several genes activated in drought.
  Goal: Analysis of beet genomes. I served as a consultant for this project for the *de novo* assembly work package.
  PI: Ali Ergül
- Scientific and Technical Research Council of Turkey (TÜBİTAK-1001-114Z927), 2015-2017
  Title: Characterization of Central Anatolian Neolithic populations by ancient DNA extraction and genome sequencing of individuals from Tepecik-Çiftlik (Niğde) and Çatalhöyük (Konya).
  Goal: Ancient DNA analysis of Neolithic populations. I served as a bioinformatics consultant for this project.
  PI: Mehmet Somel
- Scientific and Technical Research Council of Turkey (TÜBİTAK-2505-215S357), 2015-2017
  Title: Identifying and targeting sponge long non-coding RNAs to inhibit metastasis in triple-negative breast cancer using a systems biology approach.
  Goal: Delineating the mRNA-miRNA-lncRNA regulatory network controlling metastatic progression in TNBC. I served as a researcher for the bioinformatics components of this project.
  PI: Özgür Şahin
- National Institutes of Health (R01 HG006004), 2011-2016
  Title: Novel algorithms and hardware designs for ultra-fast next-gen sequence analysis.
  Goal: Develop specialized hardware architectures to accelerate mapping reads generated with the high-throughput sequencing platforms.
  PI: Onur Mutlu
  Co-PI: Can Alkan
- European Union Marie Curie Actions Career Integration Grant (PCIG10-GA-2011-303772), 2012-2016
  Title: Integrated approaches for genomic variation discovery using high throughput sequencing.
  Goal: Develop novel combinatorial algorithms to comprehensively and quickly discover genomic variation.
  PI: Can Alkan
- Scientific and Technical Research Council of Turkey (TÜBİTAK-1001-112E135), 2012-2016
  Title: Development and application of novel genome assembly algorithms that use multiple data sources.
  Goal: Develop assembly algorithms to reliably construct *de novo* genome assemblies using data from multiple sources.
  PI: Can Alkan
- National Institutes of Health (U01 HG0052209), 2009-2012
  Title: Structural Genomic Variation Analysis for the 1000 Genomes Project.
  Goal: Develop computational methods to mine structural variation data from the 1000 Genomes. As part of the consortium, I was involved in testing paired-end read approaches to detect insertions and deletions as a postdoctoral fellow at the Eichler Lab.
  PI: Charles Lee

Students and Interns (Current)

Ph.D.

- **Ricardo Román-Brenes:** Computer Engineering, Fall 2020 - present. Bilkent University.
- **Rafi Çoktalaş:** Computer Engineering, Fall 2024 - present. Bilkent University.
- **Klea Zambaku:** Computer Engineering, Fall 2025 - present. Bilkent University.
- **Akmuhammet Ashyralyyev**: Computer Engineering, Fall 2025 - present. Bilkent University.
- **Meryem Banu Cavlak**: Computer Engineering, Fall 2025 - present. Bilkent University.

M.Sc.

- **Berkan Şahin**: Computer Engineering, Fall 2024 - present. Bilkent University.
- **Arda İçöz**: Computer Engineering, Fall 2024 - present. Bilkent University.
- **Ali Erdem Karaçay**: Computer Engineering, Fall 2024 - present. Bilkent University.
- **Tuna Okçu**: Computer Engineering, Fall 2025 - present. Bilkent University.
- **Muti Kara**: Computer Engineering, Fall 2025 - present. Bilkent University.
- **Nikan Zamani**: Computer Engineering, Fall 2026 - present. Bilkent University.

Undergraduate Volunteers

- **Derin Ceylin Atalar:** B.Sc. student in Computer Engineering, Bilkent University. Since Fall 2026.
- **Yaşar Delibaş:** B.Sc. student in Computer Engineering, Bilkent University. Since Fall 2026.
- **Bilge İdil Öziş:** B.Sc. student in Computer Engineering, Bilkent University. Since Summer 2026.
- **Doğa Otçu:** B.Sc. student in Computer Engineering, Bilkent University. Since Spring 2026.
- **Ata Oğuz:** B.Sc. student in Computer Engineering, Bilkent University. Since Fall 2025.
- **Mehmet Akif Yavuz:** B.Sc. student in Computer Engineering, Bilkent University. Since Fall 2025.
- **Toygar Tanrıverdi:** B.Sc. student in Computer Engineering, Hacettepe University. Since Spring 2025.
- **Şeyma Kılıç:** B.Sc. student (now graduated) in Computer Engineering, Bilkent University. Since Fall 2023.
- **İlmay Taş:** B.Sc. student in Computer Engineering, Bilkent University. Since Fall 2024.
- **Erdem Uğurlu:** B.Sc. student in Computer Engineering, Bilkent University. Since Fall 2024.

Undergraduate Students in the Research Project Course

- **Doğa Otçu:** Lowest common ancestor algorithm using Sequence Bloom Trees. Fall 2026.
- **Ege Şeşen:** Variation graph partitioning for PIM systems. Fall 2026.
- **Emir Atlas:** Long read compression. Fall 2026.

Students and Interns (Alumni)

Postdoctoral Fellows

- **Malavi Sengupta**: Postdoc, co-advised with Ayça Ergül (Hacettepe University) and Mehmet Somel (Middle East Technical University), 2021 - 2022.
  after Bilkent: Instructor at Ashoka University, India
- **Nermin Çelik**: Postdoc, 2021 - 2023.

Ph.D.

- **Zülal Bingöl:** Ph.D., Computer Engineering, Summer 2026. Bilkent University.
  "Accelerating pangenome analysis on De Bruijn graphs"
- **Arda Söylev:** Ph.D., Computer Engineering, Fall 2018. Bilkent University.
  "Algorithms for structural variation discovery using multiple sequence signatures"
  after Bilkent: Postdoctoral Fellow at Heinrich Heine University. now: Assistant Professor at Necmettin Erbakan University.
- **Mohammed Alser:** Ph.D., Computer Engineering, Summer 2018. Bilkent University.
  Co-supervisor: Onur Mutlu.
  "Accelerating the understanding of life's code through better algorithms and hardware design"
  after Bilkent: Postdoctoral Fellow and Instructor at ETH Zürich. now: Assistant Professor at Georgia State University.

Ph.D. (as co-supervisor)

- **Mona Shojaei:** Ph.D., Bioinformatics, Fall 2023. Co-supervised with Aybar Can Acar. Middle East Technical University Informatics Institute.
  "An integrative framework for clinical diagnosis and knowledge discovery from exome sequencing data"
- **Pınar Kavak:** Ph.D., Computer Engineering, Spring 2017. Boğaziçi University. Co-supervised with Tunga Güngör.
  "Developing new approaches for multi-platform and multi-individual genomic sequence assembly"
  now: software engineer at Splunk Inc., Vancouver, BC, Canada

M.Sc.

- **Ecem İlgün:** Computer Engineering, Fall 2025. Bilkent University.
  "Pairwise Whole Genome Alignment Using Locally Consistent Parsing"
- **Ömer Yavuz Öztürk:** Computer Engineering, Summer 2025. Bilkent University.
  "Sequence-to-Graph Alignment on a Processing-In-Memory System"
  after Bilkent: software engineer at Tradeware Technology.
- **Klea Zambaku:** Computer Engineering, Summer 2025. Bilkent University.
  "Using Bloom filters to quickly and efficiently characterize genomic repeats and segmental duplications"
  now: Ph.D. student at Bilkent University.
- **Akmuhammet Ashyralyyev**: Computer Engineering, Summer 2025. Bilkent University.
  "Efficient variation graph construction using Locally Consistent Parsing"
  now: Ph.D. student at Bilkent University.
- **Rafi Çoktalaş:** Computer Engineering, Summer 2024. Bilkent University.
  "Characterization of structural variation through assembly-to-assembly comparison"
  now: Ph.D. student at Bilkent University and software engineer at CyanGate.
- **Mahmud Sami Aydın:** M.Sc., Computer Engineering, Summer 2023. Bilkent University.
  "Whole genome alignment via Alternating Lyndon Factorization Tree traversal"
  now: Ph.D. student at Stockholm University.
- **Gözde Yazıcı:** M.Sc., Computer Engineering, Fall 2022. Bilkent University.
  "Identification of protein-protein interaction bridges for multiple sclerosis"
  now: Ph.D. student at the Department of Industrial Engineering, Bilkent University.
- **Ezgi Ebren:** M.Sc., Computer Engineering, Fall 2020. Bilkent University.
  "Large structural variation discovery using long reads with several degrees of error"
  after Bilkent: research engineer at UCLA.
- **Zülal Bingöl:** M.Sc., Computer Engineering, Summer 2020. Bilkent University. Co-supervisor: Özcan Öztürk.
  "GateKeeper-GPU: accelerated pre-alignment filtering in short read mapping"
  now: Ph.D. student at Bilkent University.

- **Alim Şükrücan Gökkaya:** M.Sc., Computer Engineering, Fall 2019. Bilkent University.
  "Distributed stream-processing framework for graph-based sequence alignment"
  after Bilkent: software development engineer at Amazon London.
- **Emre Doğru:** M.Sc., Computer Engineering, Summer 2019. Bilkent University. *started the program in Spring 2016 under the supervision of Prof. Cevdet Aykanat.*
  "Paralog specific gene copy number discovery within segmental duplications"
  after Bilkent: researcher at TÜBİTAK BİLGEM.
- **Halil İbrahim Özercan:** M.Sc., Computer Engineering, Summer 2019. Bilkent University.
  "A cryptocurrency incentivized voluntary grid computing platform for DNA read alignment"
  after Bilkent: software engineer, OpsGenie. now: software engineer, Google London.
- **Balanur İçen:** M.Sc., Computer Engineering, Summer 2019. Bilkent University. *started the program in Fall 2016 under the supervision of Assoc. Prof. Selim Aksoy.*
  "Breakpoint refinement of genomic structural variation using split read analysis"
  after Bilkent: software engineer, K12Net.
- **Fatih Karaoğlanoğlu:** M.Sc., Computer Engineering, Summer 2018. Bilkent University.
  "Characterization of large structural variation using Linked-Reads"
  after Bilkent: Ph.D. at Simon Fraser University.
- **F. Tuğba Doğan:** M.Sc., Computer Engineering, Spring 2018. Bilkent University.
  "High throughput UDP-based peer-to-peer secure data transfer"
- **Can Fırtına:** M.Sc., Computer Engineering, Fall 2017. Bilkent University.
  "Assessment and correction of errors in DNA sequencing technologies"
  after Bilkent: Ph.D. at ETH Zürich. now: Assistant Professor at the University of Maryland.
- **Gülfem Demir:** M.Sc., Computer Engineering, Spring 2017. Bilkent University.
  "Characterization of short tandem repeats using local assembly"
  after Bilkent: software engineer at Seven Bridges Genomics. now: production engineer at Two Sigma.
- **Azita Nouri:** M.Sc., Computer Engineering, Spring 2016. Bilkent University.
  "Read mapping methods optimized for multiple GPGPUs"
  after Bilkent: Ph.D. at Rutgers University.
- **Marzieh Eslami Rasekh:** M.Sc., Computer Engineering, Spring 2015. Bilkent University.
  "Algorithms for the discovery of large genomic inversions using pooled clone sequencing"
  after Bilkent: Ph.D. at Boston University. now: bioinformatics scientist at Illumina.
- **Elif Dal:** M.Sc., Computer Engineering, Fall 2014. Bilkent University.
  "Genome scaffolding using pooled clone sequencing"
  after Bilkent: software engineer at TÜBİTAK BİLGEM. now: software engineer at Siemens.
- **Fatma Kahveci (née Balcı):** M.Sc., Computer Engineering, Spring 2014. Bilkent University.
  "Bias correction in finding copy number variation using read depth-based methods in exome sequencing data"
- **Mustafa Korkmaz:** M.Sc., Computer Engineering, Fall 2012. Bilkent University.
  Co-supervised with Cevdet Aykanat.
  "Massively parallel mapping of next-generation sequence reads using GPU"
  after Bilkent: Ph.D. at University of Waterloo.

Undergraduate volunteers

- **Ege Şirvan:** B.Sc. student in Computer Engineering, Bilkent University. Spring 2025 - Spring 2026.
- **Zeynep Akçil:** B.Sc. student in Electrical and Electronics Engineering, Bilkent University. Fall 2024 - Spring 2026.
- **Tuna Okçu:** B.Sc. student in Computer Engineering, Bilkent University. Fall 2024 - Spring 2025.
- **Serdar Bozdağ:** B.Sc. student in Computer Engineering, Bilkent University. Summer 2025.
- **Eslim Rana Emiroğlu:** B.Sc. student in Computer Engineering, Bilkent University. Fall 2023 - Fall 2024.

- **Mete Enis Yılmaz:** B.Sc. student in Computer Engineering, Bilkent University. Fall 2024.
- **Sıla Özel:** B.Sc. student in Computer Engineering, Bilkent University. Fall 2023 - Fall 2024.
- **Berk S Yılmaz:** B.Sc. student in Computer Engineering, Bilkent University. 2022 - 2023.
- **Oğuzhan Özçelik:** B.Sc. student in Computer Engineering, Bilkent University. Spring 2020 - Spring 2022.
- **Meryem Banu Cavlak:** B.Sc. student in Computer Engineering, Bilkent University. Summer 2019 - Spring 2021. *after Bilkent: M.Sc. at ETH Zürich. now: Ph.D. student at Bilkent.*
- **Naisila Puka:** B.Sc. student in Computer Engineering, Bilkent University. Spring 2019 - Spring 2020. *after Bilkent: software engineer at Microsoft.*
- **Aldo Tali:** B.Sc. student in Computer Engineering, Bilkent University. Fall 2019.
- **Figali Taho:** B.Sc. student in Computer Engineering, Bilkent University. Fall 2017 - Spring 2018. *after Bilkent: M.Sc. at Simon Fraser University.*
- **Baraa Orabi:** B.Sc. student in Computer Engineering, Bilkent University. Spring 2016 - Summer 2017. *after Bilkent: Ph.D. at the University of British Columbia.*
- **Ezgi Ebren:** B.Sc. student in Computer Engineering, Bilkent University. Summer 2016 - 2017.
- **Abdullah Alperen:** B.Sc. student in Computer Engineering, Bilkent University. Spring 2015 - 2017. *after Bilkent: Ph.D. student at Michigan State University.*
- **Halil İbrahim Özercan:** B.Sc. student in Computer Engineering, Bilkent University. Fall 2015 - Spring 2016.
- **Ayhun Tekat:** B.Sc. student in Computer Engineering, Bilkent University. Summer 2015.
- **Gülfem Demir:** B.Sc. student in Computer Engineering, Middle East Technical University. 2013-2014. *after Bilkent: software engineer at Seven Bridges Genomics. now: production engineer at Two Sigma.*
- **Can Koçkan:** B.Sc. student in Computer Engineering, Bilkent University. 2012-2014. *after Bilkent: Ph.D. student at Indiana University.*
- **Yiğit Küçük:** B.Sc. student in Computer Engineering, Bilkent University. 2012-2014. *after Bilkent: Ph.D. student at Case Western Reserve University.*
- **Serhat Kıyak:** B.Sc. student in Computer Engineering, Bilkent University. 2013-2014. *last seen: software engineer at Microsoft.*

Undergraduate Students in the Research Project Course

- **İlmay Taş:** Sequence to graph alignment, Spring 2026.
- **Bilge İdil Öziş:** Indexing pangenome graphs, Spring 2026.
- **Mehmet Akif Yavuz:** NPU acceleration of minimizer selection, Spring 2026.
- **Ata Soykal**: r-index for LCP cores, Fall 2025.
- **Furkan Başıbüyük**: Pangenomic extensions to the AirLift algorithm, Fall 2025.
- **Emir Tomrukçu**: NPU acceleration of the Pair-HMM algorithm, Fall 2025.
- **Bertan Uran**: NPU acceleration of the GQT algorithm, Fall 2025.
- **Şeyma Kılıç:** Suffix array-based indexing of LCP cores, Spring 2025.
- **Eslim Rana Emiroğlu:** Metagenomic read classification using raw nanopore signals, Fall 2024.
- **Begüm Filiz Öz:** Benchmarking, evaluation, and comparison of sketching techniques, Fall 2024.
- **Ege Şirvan:** Variation graph construction using Locally Consistent Parsing, Fall 2024.
- **Arda İçöz:** Benchmarking metagenomics tools. Spring 2024.
- **Şebnem Türkoğlu:** Accelerating the String Decomposer algorithm. Fall 2023.
- **Ebrar Bozkurt:** Benchmarking of exome-based copy number variation calling learner (ECOLE) tool. Spring 2023.
- **Hacı Çakın:** Analyzing genomic sequences with Bloom Filters and Counter Bloom Filters using C++. Spring 2023.
- **Ege Demirkırkan:** Structural variation discovery through genome homology maps. Fall 2022.
- **Gamze Güliter:** Benchmarking read-mapping tools for bacterial pangenomes. Spring 2022.

- **Akmuhammet Ashyralyyev:** Breakpoint refinement of structural variations with exact locations given with confidence intervals. Spring 2022.
- **Ahmet Cemal Alıcıoğlu:** Pairwise sequence alignment under block edit distance. Spring 2022.
- **Meryem Banu Cavlak:** Assembly-based genomic variation validation. Fall 2020.
- **Naisila Puka:** retroCNV discovery using high-throughput sequencing. Spring 2020.
- **Ayşe Berceste Dinçer:** Discovery of large inversions using hybrid sequencing technologies, Fall 2016.
- **Can Fırtına:** Fast filters for pairwise comparison of short and long NGS reads. Spring 2015.
- **Damla Şenol:** Transmission patterns discovery of genome structural variations among generations of a family. Spring 2015.
- **Atalay Mert İleri:** ScienceCoin: a cryptocurrency with DNA sequence alignment as proof-of-work. Spring 2014.

Interns

- **Toygar Tanrıverdi:** B.Sc. student in Computer Engineering, Hacettepe University. Summer 2025.
- **Murat Eren Güven:** B.Sc. student in Computer Engineering, Hacettepe University. January-February, 2025.
- **Elif Turan:** Intern, Summer 2022. *from: İzmir University of Economics.*
- **Fatemeh Hasiri:** Intern, Summer 2017 at Bilkent. *from: Amirkabir University of Technology.*
- **Farnush Farhadi:** Intern, Summer 2014. *from: Sharif University.*
- **Vineet Bhakhar:** Intern, Summer 2014. *from: IIT Guwahati.*

Others

- **Ebrar Bozkurt**: M.Sc. student, Computer Engineering, Fall 2023 - 2026. Bilkent University. Left the program in Spring 2026.
- **Fatma Kahveci:** Ph.D. candidate, Computer Engineering, Fall 2014 - 2021. Bilkent University. Left the program in Fall 2021.
- **Mete Han Kahraman:** M.Sc. student, Computer Engineering, Fall 2019. Bilkent University. Left the program in Spring 2020.
- **Shatlyk Asyralyyev:** Ph.D. student, Computer Engineering, Fall 2013 - 2016. Bilkent University. Co-supervised with Cevdet Aykanat. Left the program in Fall 2016.
- **Dilek Genç:** M.Sc. student, Computer Engineering, Fall 2015. Bilkent University. Left the program in Spring 2016.
- **Farhad Hormozdiari:** Research Scientist at the Eichler Lab between February and September 2010. *last seen: Google Health.*
- **Sònia Casillas:** Visiting scientist at the Eichler Lab between February and July 2011.
- **Michael Duyzend:** Rotation student at the Eichler Lab between June and August 2010. *last seen: Resident Physician at Boston Children's Hospital.*
- **Pietro D'Addabbo:** Visiting scientist at the Eichler Lab between August and November 2009. *last seen: Staff scientist at University of Bari.*
- **Peter H. Sudmant:** Ph.D. student at the Eichler Lab between May 2009 and December 2011. *now: Assistant Professor at University of California, Berkeley.*
- **Jacob O. Kitzman:** Rotation student at the Eichler Lab between January and April 2009. *now: Associate Professor at University of Michigan.*
- **Iman Hajirasouliha:** Intern UW between February to July 2009 at UW. *now: Associate Professor at Weill Cornell Graduate School of Medical Sciences.*
- **Fereydoun Hormozdiari:** Intern between February to July 2008 at UW. *now: Associate Professor at the University of California, Davis.*
- **Kerry Hall:** Intern at UW in 2006.

Thesis Committees (Current)

Ph.D.

- **Muazzez Çelebi**: Ph.D., Molecular Biology and Genetics, Spring 2025 - present. Advisor: Özlen Konu. Bilkent University.

Thesis Committees (Past)

Ph.D.

- **Nika Mansouri Ghiasi:** Ph.D., Information Technology and Electrical Engineering, Spring 2026. Advisor: Onur Mutlu. ETH Zurich.
- **Halil İbrahim Kuru**: Ph.D., Computer Engineering, Spring 2025. Advisors: Ercüment Çiçek, Öznur Taştan. Bilkent University.
- **Lukas Breitwieser:** Ph.D., Information Technology and Electrical Engineering, Fall 2024. Advisor: Onur Mutlu. ETH Zurich.
- **Alperen Dalkıran:** Ph.D., Computer Engineering, Summer 2024. Advisor: Volkan Atalay. Middle East Technical University.
- **Dilek Koptekin**: Ph.D., Molecular Biology and Genetics, Summer 2022. Advisor: Mehmet Somel. Middle East Technical University.
- **Damla Şenol Çalı:** Ph.D., Computer Engineering, Summer 2021. Advisor: Onur Mutlu, Saugata Ghose. Carnegie Mellon University.
- **Nour M.N. Alserr:** Ph.D., Computer Engineering, Summer 2020. Advisors: Özgür Ulusoy, Erman Ayday. Bilkent University.
- **Gülden Olgun:** Ph.D., Computer Engineering, Summer 2019. Advisor: Öznur Taştan, Ercüment Çiçek. Bilkent University.
- **Fahrettin Can Koyuncu:** Ph.D., Computer Engineering, Fall 2018. Advisor: Çiğdem Gündüz Demir. Bilkent University.
- **Abdullah Atmaca:** Ph.D., Computer Engineering, Summer 2018. Advisors: Yavuz Oruç and Cevdet Aykanat. Bilkent University.
- **Arif Yılmaz:** Ph.D., Bioinformatics, Spring 2018. Advisor: Yeşim Aydın Son. Middle East Technical University.
- **Burçak Otlu:** Ph.D., Computer Engineering, Spring 2017. Advisor: Tolga Can. Middle East Technical University.
- **Evrim Tekeli:** Ph.D., Anthropology, Spring 2017. Advisor: Timur Gültekin. Ankara University.
- **Onur Baloğlu:** Ph.D. candidate, Biology, since Fall 2015 - Spring 2018. I left the committee in Spring 2018. Advisor: Mehmet Somel. Middle East Technical University.
- **Daria Iakovishina:** Ph.D., Bioinformatics, November 2015. Advisor: Valentina Boeva. École Polytechnique, Palaiseau.
- **Yogesh Paudel:** Ph.D., Biostatistics and Informatics, January 2015. Advisor: Martien A.M. Groenen. Institut für Biostatistik und Informatik in Medizin und Alternsforschung, Wageningen University, Wageningen.
- **Gülşah Merve Dal:** Ph.D., Molecular Biology and Genetics, Fall 2014. Advisor: Tayfun Özçelik. Bilkent University.
- **Özlem Tufanlı:** Ph.D. candidate, Molecular Biology and Genetics, since Fall 2013 - Spring 2015. I left the committee in Spring 2015. Advisor: Ebru Erbay. Bilkent University.

M.Sc.

- **Sobhan Shokoueiantabrizi:** M.Sc., Computer Engineering, Fall 2026. Advisor: Ercüment Çiçek. Bilkent University.
- **Mehmet Alper Yılmaz:** M.Sc., Computer Engineering, Spring 2026. Advisor: Ercüment Çiçek. Bilkent University.
- **Muammer Buğra Kurnaz:** M.Sc., Computer Engineering, Spring 2026. Advisor: Ercüment Çiçek. Bilkent University.
- **Masoud Poorghaffar Aghdam:** M.Sc., Computer Engineering, Fall 2025. Advisor: Ercüment Çiçek. Bilkent University.

- **Berk Çiçek:** M.Sc., Computer Engineering, Summer 2025. Advisor: Özgür S. Öğüz. Bilkent University.
- **Barış Bilgin Şenol:** M.Sc., Computer Engineering, Summer 2025. Advisor: Özgür S. Öğüz. Bilkent University.
- **Melih Coşğun:** M.Sc., Computer Engineering, Summer 2025. Advisor: Sinem Sav. Bilkent University.
- **Yahya Elnouby:** M.Sc., Computer Engineering, Summer 2025. Advisor: Eray Tüzün. Bilkent University.
- **Sina Barazandeh:** M.Sc., Computer Engineering, Summer 2024. Advisor: Ercüment Çiçek. Bilkent University.
- **Gün Kaynar:** M.Sc., Computer Engineering, Summer 2024. Advisor: Ercüment Çiçek. Bilkent University.
- **Haya Shamim Khan Khattak:** M.Sc., Computer Engineering, Fall 2022. Advisor: Çiğdem Gündüz Demir. Bilkent University.
- **Mubashira Zaman:** M.Sc., Computer Engineering, Fall 2021. Advisor: Uğur Doğrusöz. Bilkent University.
- **Kerem Ayöz:** M.Sc., Computer Engineering, Summer 2021. Advisor: Ercüment Çiçek. Bilkent University.
- **Doruk Çakmakçı:** M.Sc., Computer Engineering, Summer 2021. Advisor: Ercüment Çiçek. Bilkent University.
- **Furkan Özden:** M.Sc., Computer Engineering, Summer 2021. Advisor: Ercüment Çiçek. Bilkent University.
- **Mustafa Duymuş:** M.Sc., Computer Engineering, Spring 2021. Advisor: Cevdet Aykanat. Bilkent University.
- **Furkan Mustafa Akdemir:** M.Sc., Computer Engineering, Fall 2020. Advisors: Öznur Taştan, Ercüment Çiçek.
- **İlayda Beyreli:** M.Sc., Computer Engineering, Fall 2020. Advisor: Ercüment Çiçek.
- **Oğuzhan Karakahya:** M.Sc., Computer Engineering, Fall 2020. Advisor: Ercüment Çiçek.
- **Emin Onur Karakaşlar:** M.Sc., Computer Engineering, Summer 2020. Advisor: Ercüment Çiçek. Bilkent University.
- **Alper Eroğlu:** M.Sc., Computer Engineering, Summer 2020. Advisor: Ercüment Çiçek. Bilkent University.
- **Muhammed Çavuşoğlu:** M.Sc., Computer Engineering, Summer 2020. Advisor: Cevdet Aykanat. Bilkent University.
- **Simla Burcu Harma:** M.Sc., Computer Engineering, Fall 2019. Advisor: Oğuz Ergin. TOBB University of Economics and Technology.
- **Zahit Saygın Doğu:** M.Sc., Computer Engineering, Summer 2019. Advisor: Cevdet Aykanat. Bilkent University.
- **Gizem Çaylak:** M.Sc., Computer Engineering, Summer 2019. Advisor: Ercüment Çiçek. Bilkent University.
- **Halil İbrahim Kuru:** M.Sc., Computer Engineering, Spring 2019. Advisor: Öznur Taştan, Ercüment Çiçek. Bilkent University.
- **Simge Yücel:** M.Sc., Computer Engineering, Fall 2018. Advisor: Çiğdem Gündüz Demir. Bilkent University.
- **M. Yağmur Şahin:** M.Sc. Computer Engineering, Fall 2016. Advisor: Buğra Gedik. Bilkent University.
- **Mona Shojaei:** M.Sc. Bioinformatics, Summer 2016. Advisor: Rengül Çetin Atalay. Middle East Technical University Informatics Institute.
- **Melike Dönertaş:** M.Sc. Biology, Spring 2016. Advisor: Mehmet Somel. Middle East Technical University.
- **Doğukan Çağatay:** M.Sc. Computer Engineering, Fall 2015. Advisor: Buğra Gedik. Bilkent University.

- **Gülsüm Ece Ekşi:** M.Sc. Computer Engineering, Fall 2015. Advisor: Bedir Tekinerdoğan. Bilkent University.
- **Onur Özdemir:** M.Sc. Computer Engineering, Fall 2015. Advisor: Bedir Tekinerdoğan. Bilkent University.
- **Havva Gülay Gürbüz:** M.Sc. Computer Engineering, Summer 2014. Advisor: Bedir Tekinerdoğan. Bilkent University.
- **Mehmet Başaran:** M.Sc. Computer Engineering, Fall 2014. Advisor: Cevdet Aykanat. Bilkent University.
- **Merve Çakır:** M.Sc. Computer Engineering, Spring 2013. Advisor: Uğur Doğrusöz. Bilkent University.
- **Fahrettin Can Koyuncu:** M.Sc. Computer Engineering, Fall 2012. Advisor: Çiğdem Gündüz Demir. Bilkent University.

Academic Service

University Duties

- Vice Chair, Department of Computer Engineering, since September 2025.
- Academic Coordinator for Erasmus+ and International Exchange Programs, Department of Computer Engineering, Bilkent University, since Summer 2014.
- Member of the Faculty Board, Bilkent University, Fall 2021 - present.
- Member of the Faculty Executive Board, Bilkent University, Fall 2021 - Fall 2026.
- CS491 Senior Design Project, Project Evaluator, Fall 2020.

Editorial & Program Committees

- **Associate Editor**, Bioinformatics, 2021-*present*.
- **Member of the Editorial Board**, Genome Research, 2020-*present*.
- **Steering Committee Member**, RECOMB Conference on Biological Sequence Analysis (RECOMB-Seq), 2017-*present*.
- **Steering Committee Member**, RECOMB Conference on Hardware Acceleration of Bioinformatics Workloads (RECOMB-Arch), 2025-*present*.
- **General Co-Chair**, 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB), 2023.
- **Organization Committee Member**, ISCB COSI for High Throughput Sequencing and Applications (HiTSeq), 2016-*present*.
- **Organization Committee Member**, Computational Genomics Summer Institute (CGSI), 2026.
- **Associate Editor**, BMC Bioinformatics, 2010-2021.
- **Track and Area Co-Chair**, Annual International Conference on Intelligent Systems for Molecular Biology and the European Conference on Computational Biology.
  - ISMB: 2018 (Chicago, IL, USA), 2016 (Orlando, FL, USA)
  - ISMB-ECCB: 2021 (Lyon, France), 2019 (Basel, Switzerland), 2017 (Prague, Czech Republic).
- **Program Committee Chair**, 21st IEEE International Workshop on High-Performance Computational Biology (HiCOMB 2022), May 30, 2022, Lyon, France (held virtually).
- **PC Member**, IEEE International Workshop on High-Performance Computational Biology
  - HiCOMB 2025 (Milan, Italy).
- **Program Committee Chair**, First Annual RECOMB Workshop on Massively Parallel Sequencing (RECOMB-seq), March 26-27, 2011, Vancouver, BC, Canada.

- **PC Member**, Annual International Conference on Research in Computational Molecular Biology
  - ◇ RECOMB: 2027 (Toronto, ON, Canada), 2026 (Thessaloniki, Greece), 2025 (Seoul, South Korea), 2024 (Cambridge, MA, USA), 2023 (İstanbul, Turkey), 2022 (San Diego, CA, USA), 2021 (Padova, Italy), 2020 (Padova, Italy), 2019 (Washington, DC, USA), 2016 (Santa Monica, CA, USA), 2011 (Vancouver, BC, Canada).
- **PC Member**, Annual RECOMB Workshop on Biological Sequence Analysis
  - ◇ RECOMB-Seq: 2024 (Cambridge, MA, USA), 2018 (Paris, France), 2017 (Hong Kong), 2013 (Beijing, China), 2012 (Barcelona, Spain).
- **PC Member**, Annual International Conference on Intelligent Systems for Molecular Biology and European Conference on Computational Biology
  - ◇ ISMB: 2024 (Montreal, Canada), 2020 (online), ISMB 2014 (Boston, MA, USA), 2012 (Long Beach, CA, USA), 2010 (Boston, MA, USA).
  - ◇ ISMB-ECCB: 2015 (Dublin, Ireland), 2013 (Berlin, Germany), 2011 (Vienna, Austria).
  - ◇ ECCB: 2026 (Geneva, Switzerland)
- **PC Member**, Conference on High Throughput Sequencing Analysis and Algorithms
  - ◇ HiTSeq 2015 (Dublin, Ireland).
- **PC Member**, HPCA Workshop on Accelerator Architecture in Computational Biology and Bioinformatics
  - ◇ AACBB 2018 (Vienna, Austria).
- **PC Member**, ACM Conference on Bioinformatics, Computational Biology, and Health Informatics
  - ◇ ACM-BCB: 2022 (Chicago, IL, USA), 2021 (Virtual), 2020 (Virtual), 2019 (Niagara Falls, NY, USA), 2017 (Boston, MA, USA).
- **PC Member**, International Symposium on Health Informatics and Bioinformatics
  - ◇ HIBIT: 2022 (Silifke, Turkey), 2021 (Ankara, Turkey), 2020 (Tuzla, Turkey), 2017 (Güzelyurt, Northern Cyprus), 2013 (Ankara, Turkey).
- **PC Member**, Next Generation Sequencing Conference
  - ◇ NGS 2017 (Barcelona, Spain).
- **PC Member**, International Conference on Algorithms for Computational Biology
  - ◇ AlCoB: 2021 (Virtual), 2019 (Berkeley, CA, USA), 2017 (Aveiro, Portugal), 2016 (Trujillo, Spain).
- **Track Chair**, 12th National Medical Genetics Congress, Çeşme, Turkey, October 7, 2016.
- **Co-Chair**, Bertinoro Computational Biology Meeting, June 14-17, 2015, Bertinoro, Italy.
- **Track Co-Chair**, Session on Personal Genomics at the Pacific Symposium for Biocomputing
  - ◇ PSB: 2011 (Big Island of Hawaii, USA), 2010 (Big Island of Hawaii, USA).
- **PC Member**, The IEEE International Conference on Advanced Information Networking and Applications
  - ◇ AINA: 2008 (Okinawa, Japan), 2007 (Niagara Falls, Canada).

Scientific Advisory Boards

- Seven Bridges Genomics, Cambridge, MA, United States. 2016 - 2018.
- Prototype Capital, Los Angeles, CA, United States. 2018 - 2020.
- Turkish Institutes of Health (TÜSEB) Genome Project, Ankara, Turkey. 2016 -2019.
- Center for Advanced Genomics and Bioinformatics (İGBAM) in TÜBİTAK Marmara Research Center, Gebze, Turkey. 2012 - 2014.

Consortium Memberships

- 1000 Genomes Project Structural Variation Analysis Group, 2008-2015, 2019-2021.
- Genome in a Bottle Consortium, 2015-*present*
- Bonobo Genome Consortium, 2008-2012.
- Orangutan Genome Consortium, 2008-2011.
- Neandertal Genome Consortium, 2009-2012.
- Gorilla Genome Consortium, 2009-2013.
- Denisovan Genome Consortium, 2010-2012.
- Great Ape Diversity Project, 2011-2014.
- Global Alliance for Genomic Health. November 2014 - 2016.

Ad-hoc Reviewer

- Nature Genetics, Nucleic Acids Research, Nature Reviews Genetics, Genome Research, PLoS Genetics, PLoS Computational Biology, Nature Microbiology, Nature Biotechnology, BMC Genomics, Genome Biology, Methods, Transactions of Computational Biology and Bioinformatics, Bioinformatics, Briefings in Bioinformatics, RECOMB, RECOMB-seq, SODA, FOCS, ICDE, CPM, AINA, PSB, ISMB, HiTSeq.

Teaching

- CS 476 Automata Theory and Formal Languages: Fall 2014, 2015, 2016, 2017, 2022, 2023, 2024, 2025, 2026; Spring 2014, 2015, 2016, 2017, 2019, 2020, 2021, 2022.
- CS 481 Bioinformatics Algorithms: Fall 2012, 2013, 2016, 2017, 2019, 2020, 2021, 2022; Spring 2024, 2025, 2026.
- CS 681 Advanced Topics in Computational Biology: Spring 2012, 2013; Fall 2019, 2020.
- CS 202 Fundamental Structures of Computer Science II: Spring 2013, 2023, 2024, 2025, 2026; Fall 2021, 2023, 2024, 2025.
- CS 590/690 Research Topics (coordinator): Spring 2012, 2014, 2019, 2022.
- CS 319 Object-Oriented Software Engineering: Summer 2012; Spring 2013; Fall 2013, 2014, 2015.

Senior Project Supervision

- Genie5: Literature-derived gene regulatory network builder, 2026-2027
- CASSIE: Cloud-based Assembly Streamlined Service Integration Engine, 2025-2026.
- Pathogenious: Pathogen identification using edge computing, 2025-2026.
- Conferencer: Conference management system, 2024-2025.
- YOLLA: Package delivery marketplace, 2024-2025.
- TradeBlocs: Crypto-trading platform, 2024-2025.
- Evalio: Exam database and management system, 2024-2025.
- RestoRun: Restaurant automation application, 2023-2024.
- Block Script: An educational visual programming language, 2023-2024.
- SoCoolBus: Student school bus tracking system, 2022-2023.
- Vanny: Virtual nanny, 2022-2023.
- Neutlan: NLP-based detection of gender discrimination in texts, 2022-2023.
- Tutorium: Online tutoring system, 2022-2023.
- Cassett: Bicyclist assistance tool, 2021-2022.
- ShopEye: Personal assistant for visually impaired people, 2021-2022.
- Laber: Real-time social media analysis, 2021-2022.
- onRoute: A road trip assistance tool, 2020-2021.
- LIBRA: Integrated web-based system to aid diagnosis using clinical sequencing, 2019-2020.

- Pengout: Mobile application for finding events and organizing group outings, 2019.
- CryptDist: Content distribution network using the blockchain technology, 2016-2017.
- Espionage Game: Stealth game with adaptive artificial intelligence, 2015-2016.
- BioPeer: Secure research data sharing platform, 2014-2015.
- Coinami: Cryptocurrency mining system for scientific computation, 2014-2015.
- Anotice: Anonymized social network platform, 2014-2015.
- Touravel: Travel logging application, 2014-2015.
- SUN (Social University Network): Study group game application, 2014-2015.
- SmartVote: Electronic Voting System for Mobile Devices, 2013-2014.
  Recipient of the Best Senior Project Award in 2014.
- Daycept: Concept-based idea-sharing platform, 2013-2014.
- Betcha: Social platform that enables betting among friends, 2013-2014.
- SocioDiary: Digital diary system, 2013-2014.
- CMPS101: Activity planning for university students, 2013-2014.
- chkDNA: Genomic variation analysis tool, 2012-2013.

Other Service

- Member, ISCB Awards Committee, 2021 - present
- Observer to ELIXIR for Turkey. 2013 - 2015.

Selected Conference and Invited Talks

A total of 78 invited talks, seminars, lectures, and keynotes.

- Invited Speaker, **String processing with Locally Consistent Parsing and applications to pangenomics.** Computational Genomics Summer Institute, University of California, Los Angeles, CA, United States, July 14, 2026.
- Invited Speaker, **Acceleration of bioinformatics workloads through hardware/algorithm co-design.** Vancouver Bioinformatics Users Group (VanBUG), Vancouver, BC, Canada (online). November 21, 2024.
- Invited Speaker, **Acceleration of bioinformatics workloads through hardware/algorithm co-design and processing-in-memory technologies.** Computational Genomics Summer Institute, University of California, Los Angeles, CA, United States, July 16, 2024.
- Invited Speaker, **Algorithms to characterize genomic structural variation using high throughput sequencing technologies.** Personalized Medicine and Functional Genomics Meeting, Sidra Medicine, Doha, Qatar, September 26, 2022.
- Invited Speaker, **Acceleration of read mapping through hardware/software co-design.** Computability in Europe 2020, Salerno, Italy, June 30, 2020.
- Lecture, **Characterization of genomic structural variation using various sequencing technologies.** Computational Genomics Summer Institute, University of California, Los Angeles, CA, United States, July 19, 2019.
- Invited Speaker, **Population reference graphs – a primer to computational pan-genomics.** Clinical genomics and NGS hybrid course, European School of Genetic Medicine and Acıbadem University, İstanbul, Turkey, April 30, 2019.
- Invited Speaker, **Addressing computational burden for low-priority genome analyses.** HPCA 2018 Workshop On Accelerator Architecture in Computational Biology and Bioinformatics, Vienna, Austria, February 24, 2018.
- Instructor, **EMBO Practical Course: Computational biology: from genomes to cells and systems.** Cappadocia, Nevşehir, Turkey. September 29 - October 4, 2013.
- Invited Speaker, **Characterization of mobile element insertions using high throughput sequencing.** 63rd Fujihara Seminar: A new horizon of retroposon research, Kyoto, Japan, August 2, 2012.

- Invited Speaker, **Next-generation sequence characterization of complex genome structural variation.** Department of Genetics, Yale University, New Haven, CT, USA, May 11, 2011.
- Invited Speaker, **Dark side of genomes: what is missing in current sequence assemblies?** Genome 10K Project Genome Assembly Workshop, Santa Cruz, CA, USA, March 16, 2011.
- Keynote Speaker, **Structural variation discovery and characterization of segmental Duplications with next-gen sequencing technologies**. Spanish and Portuguese Bioinformatics Conference (Jornadas de Bioinformática), Lisbon, Portugal, November 6, 2009.
- Conference Talk, **RNA-RNA interaction prediction and antisense RNA target search**. Ninth Annual Conference on Research in Computational Molecular Biology (RECOMB 2005), Cambridge, MA, USA, May 14, 2005.

Publications

Journal Publications

Joint first authors are marked with $*$ in case of equal contribution. Joint last authors are marked with ‡.

De Bruijn graphs for pangenomics: in-depth performance benchmarking of de Bruijn graph-based tools for read mapping. Z Bingöl, B Şahin, K Zambaku, R Roman-Brenes, K Koliogeorgi, C Firtina, O Mutlu, **C Alkan**. *Briefings in Bioinformatics*, 27 (4): bbag440, 2026.

PG2: algorithms and a web-based tool for effective layout and visual analysis of pangenome graphs. GK Solun, U Dogrusoz, Z Bingöl, **C Alkan**. *BMC Genomics*, 27: 236, 2026.

LCPan: efficient variation graph construction using Locally Consistent Parsing. A Ashyralyyev, Z Bingöl, BF Öz, K Zhu, S Malikic, U Vishkin, SC Sahinalp, **C Alkan**. *Genome Biology*, 27: 197, 2026. *Preprint: arXiv:2511.12205.*

Rapid adaptive increase of amylase gene copy number in indigenous Andeans. K Scheer, LJB Landau, K Jorgensen, C Karageorgiou, L Siao, **C Alkan**, AM Morales-Rivera, C Osbourne, O Garcia, L Pearson, M Kiyamu, M Rivera-Chira, F Leon-Velarde, F Lee, T Brutsaert, AW Bigham, O Gokcumen. *Nature Communications*, 17: 3822, 2026. *Preprint: bioRxiv 2025.03.25.644684.*

AirLift: a fast and comprehensive technique for translating alignments between reference genomes. JS Kim, C Firtina, D Senol Cali, N Hajinazar, M Alser, **C Alkan**‡, O Mutlu‡. *IEEE Transactions on Computational Biology and Bioinformatics*, 23 (2): 589-597, 2026. *Preprint: arXiv:1912.08735.*

presented at the 21st Asia Pacific Bioinformatics Conference (APBC 2023), Changsha, Hunan, China, April 14-16, 2023.

Processing-in-memory for genomics workloads. WA Simon, L Yavits, K Koliogeorgi, Y Falevoz, Y Shibuya, D Lavenier, I Boybat, K Zambaku, B Şahin, M Sadrosadati, O Mutlu, A Sebastian, R Chikhi, The BioPIM Consortium, **C Alkan**. *IEEE Micro*, 46 (2): 70-80, 2026. *Preprint: doi: 10.48550/arXiv.2506.00597.*

Population-level gene copy number variations reveal distinct genetic properties of different *Malus* species. A Sakina, **C Alkan**, A Khan. *BMC Genomics*, 26: 687, 2025.

Pre-processing of paleogenomes: Mitigating reference bias and postmortem damage in ancient genome data. D Koptekin, E Yapar, KB Vural, E Sağlıcan, NE Altınışık, AS Malaspinas, **C Alkan**, M Somel. *Genome Biology*, Jan 9; 26: 6, 2025. *Preprint: bioRxiv 2023.11.11.566695.*

TargetCall: eliminating the wasted computation in basecalling via pre-basecalling filtering. MB Cavlak, G Singh, M Alser, C Firtina, J Lindegger, M Sadrosadati, N Mansouri Ghiasi, **C Alkan**, O Mutlu. *Frontiers in Genetics*, 15, 2024. *Preprint: arXiv:2212.04953.*

presented at the 21st Asia Pacific Bioinformatics Conference (APBC 2023), Changsha, Hunan, China, April 14-16, 2023.

GateKeeper-GPU: fast and accurate pre-alignment filtering in short read mapping. Z Bingöl, M Alser, O Mutlu, O Ozturk, **C Alkan**. *IEEE Transactions on Computers*, 73 (5): 1206-1218, 2024. *Preprint: arXiv:2103.14978.*

An integrative framework for clinical diagnosis and knowledge discovery from exome sequencing data. M Shojaei, N Mohammadvand, T Doğan, **C Alkan**, R Çetin Atalay, AC Acar. *Computers in Biology and Medicine*, 169: 107810, 2024.

ECOLE: Learning to call copy number variants on whole exome sequencing data. B Mandiracioglu*, F Ozden*, G Kaynar, MA Yılmaz, **C Alkan**, AE Cicek. *Nature Communications*, 15: 132, 2024. *Preprint: bioRxiv 2022.11.17.516880.*

Identification of protein-protein interaction bridges for multiple sclerosis. G Yazıcı, B Kurt Vatandaslar, I Aydin Canturk, FI Aydinli, O Arici Duz, E Karakoc, BE Kerman, **C Alkan**. *Bioinformatics*, 39 (4): btad175, 2023.

BLEND: a fast, memory-efficient and accurate mechanism to find fuzzy seed matches in genome analysis. C Firtina, J Park, JS Kim, M Alser, D Senol Cali, T Shahroodi, N Mansouri-Ghiasi, G Singh, K Kanellopoulos, **C Alkan**, O Mutlu. *NAR Genomics and Bioinformatics*, 5 (1): lqad004, 2023. *Preprint: arXiv:2112.08687.*

CONGA: copy number variation genotyping in ancient genomes and low-coverage sequencing data. A Söylev, SS Çokoglu, D Koptekin, **C Alkan**, M Somel. *PLoS Comput Biol.* 18 (12): e1010788, 2022. *Preprint: bioRxiv 2021.12.17.473150.*

High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios. M Byrska-Bishop, US Evani, X Zhao, AO Basile, HJ Abel, AA Regier, A Corvelo, WE Clarke, R Musunuri, K Nagulapalli, S Fairley, A Runnels, L Winterkorn, E Lowy, **Human Genome**

Structural Variation Consortium, P Flicek, S Germer, H Brand, IM Hall, ME Talkowski, G Narzisi, MC Zody. *Cell*, 185 (18): 3426-3440, 2022. *Preprint: bioRxiv 2021.02.06.430068.*

C Alkan *is a member of The Human Genome Structural Variation Consortium.*

FastRemap: a tool for quickly remapping reads between genome assemblies. JS Kim, C Firtina, MB Cavlak, D Senol Cali, **C Alkan**, O Mutlu. *Bioinformatics*, 38 (19): 4633-4635, 2022. *Preprint: arXiv:2201.06255.*

Polishing copy number variant calls on exome sequencing data via deep learning. F Özden, **C Alkan**‡, AE Çiçek‡. *Genome Research*, 32: 1170-1182, 2022. *Preprint: bioRxiv 2020.05.09.086082.*

***Commentary*: Implications of the first complete human genome assembly. C Alkan**, L Carbone, M Dennis, J Ernst, G Evrony, S Girirajan, D Chi, Y Leung, CCY Cheng, D MacAlpine, T Ni, M Ramsay, H Rowe, P Gould, R Enriquez-Gasca, B Sullivan. *Genome Research*, 32 (4): 595-598, 2022.

Fast characterization of segmental duplication structure in multiple genome assemblies. H Išerić, **C Alkan**, F Hach, I Numanagić. *Algorithms Mol Biol.*, Mar 18; 17(1):4, 2022.

presented at the 21st International Workshop on Algorithms in Bioinformatics (WABI 2021), held virtually.

Technology dictates algorithms: recent developments in read alignment. M Alser, J Rotman, D Deshpande, K Taraszka, H Shi, P Icer Baykal, HT Yang, V Xue, S Knyazev, BD Singer, B Balliu, D Koslicki, P Skums, A Zelikovsky, **C Alkan**, O Mutlu, S Mangul. *Genome Biology*, Aug 26; 22: 249, 2021. *Preprint: arXiv:2003.00110.*

SneakySnake: a fast and accurate universal genome pre-alignment filter for CPUs, GPUs, and FPGAs. M Alser, T Shahroodi, J Gomez-Luna, **C Alkan**‡, O Mutlu‡. *Bioinformatics*, 36 (22-23), 5282–5290, 2020. *Preprint: arXiv:1910.09020.*

Accelerating Genome Analysis: a primer on an ongoing journey. M Alser, Z Bingöl, J Kim, D Senol Cali, S Ghose, **C Alkan**, O Mutlu. *IEEE Micro*, 40 (5): 65-75, 2020. *Preprint: arXiv:2008.00961.*

A robust benchmark for detection of germline large deletions and insertions. JM Zook, NF Hansen, ND Olson, LM Chapman, JC Mullikin, C Xiao, S Sherry, S Koren, AM Phillippy, PC Boutros, SME Sahraeian, V Huang, A Rouette, N Alexander, CE Mason, I Hajirasouliha, C Ricketts, J Lee, R Tearle, IT Fiddes, A Martinez-Barrio, J Wala, A Carroll, N Ghaffari, OL Rodriguez, A Bashir, S Jackman, JJ Farrell, AM Wenger, **C Alkan**, A Soylev, MC Schatz, S Garg, G Church, T Marschall, K Chen, X Fan, AC English, JA Rosenfeld, W Zhou, RE Mills, JM Sage, JR Davis, MD Kaiser, JS Oliver, AP Catalano, MJP Chaisson, N Spies, FJ Sedlazeck, M Salit. *Nature Biotechnology*, Nov; 38 (11):1347-1355, 2020. *Preprint: bioRxiv 664623.*

Apollo: a sequencing-technology-independent, scalable, and accurate assembly polishing algorithm. C Firtina, JS Kim, M Alser, D Senol Cali, AE Cicek, **C Alkan**‡, O Mutlu‡. *Bioinformatics*, 36 (12): 3669-3679, 2020. *Preprint: arXiv:1902.04341.*

VALOR2: characterization of large-scale structural variants using linked-reads. F Karaoğlanoğlu*, C Ricketts*, E Ebren, M Eslami Rasekh, I Hajirasouliha‡, **C Alkan**‡. *Genome Biology*, Mar 19; 21: 72, 2020. *Preprint: bioRxiv 394528.*

Automatic characterization of copy number polymorphism using high throughput sequencing. **C Alkan**. *Turkish J Elec Eng & Comp Sci*, 28: 253-261, 2020.

Shouji: a fast and efficient computation of banded sequence alignment. M Alser, H Hassan, A Kumar, O Mutlu‡, **C Alkan**‡. *Bioinformatics*, Nov 1; 35 (21): 4255–4263, 2019. *Preprint: arXiv:1809.07858.*

Discovery of tandem and interspersed segmental duplications using high-throughput sequencing. A Soylev*, T Le*, H Amini, **C Alkan**‡, F Hormozdiari‡. *Bioinformatics*, Oct 15; 35(20): 3923-3930, 2019. *Preprint: bioRxiv 393694.*

Nanopore sequencing technology and tools for genome assembly: computational analysis of the current state, bottlenecks and future directions. D Senol Cali, J Kim, S Ghose, **C Alkan**‡, O Mutlu‡. *Briefings in Bioinformatics*, [epub Apr 2, 2018], Jul; 20(4): 1542–1559, 2019. *Preprint: arXiv:1711.08774.*

Characterizing microsatellite polymorphisms using assembly-based and mapping-based tools. G Demir and **C Alkan**. *Turkish J Biol*, Aug; 43 (4): 264-273, 2019.

Evaluation of genome scaffolding tools using pooled clone sequencing. E Dal and **C Alkan**. *Turkish J Biol*, Dec; 42 (6): 471-476, 2018.

Fast characterization of segmental duplications in genome assemblies. I Numanagić, AS Gökkaya, L Zhang, B Berger, **C Alkan**‡, F Hach‡. *Bioinformatics*, Sep 1; 34(17): i706-714, 2018. *Preprint: arXiv:1807.00205.*

presented at the 17th European Conference for Computational Biology (ECCB 2018), September 8-12, 2018, Athens, Greece.

Hercules: a profile HMM-based hybrid error correction algorithm for long reads. C Firtina, Z Bar-Joseph, **C Alkan**‡, AE Cicek‡. *Nucleic Acids Research*, Nov 30; 46(21): e125, 2018. *Preprint: bioRxiv 233080.*

Realizing the potential of blockchain technologies in genomics. HI Ozercan, AM Ileri, E Ayday, **C Alkan**. *Genome Research*, Sep; 28(9): 1255-1263, 2018.

GRIM-Filter: fast seed location filtering in DNA read mapping using processing-in-memory technologies. J Kim, D Senol Cali, H Xin, D Lee, S Ghose, M Alser, H Hassan, O Ergin, **C Alkan**‡, O Mutlu‡. *BMC Genomics*, May; 19 (Suppl 2): 89, 2018. *Preprint: arXiv:1711.01177.*

presented at the 16th Asia Pacific Bioinformatics Conference (APBC 2018), January 15-17, 2018, Yokohama, Japan.

Targeting PLK1 overcomes T-DM1 resistance via CDK1-dependent phosphorylation and inactivation of Bcl-2/xL in HER2-positive breast cancer. Ö. Saatci, S Borgoni, Ö. Akbulut, S Durmuş, U Raza, E Eyüpoğlu, **C Alkan**, A Akyol, Ö. Kütük, S Wiemann, Ö. Şahin. *Oncogene*, Apr; 37(17):2251-2269, 2018.

Computational pan-genomics: status, promises and challenges. Computational Pan-Genomics Consortium. *Briefings in Bioinformatics*, [epub Oct 21, 2016], Jan 1; 19 (1): 118-135, 2018. *Preprint: bioRxiv 043430.*

GateKeeper: a new hardware architecture for accelerating pre-alignment in DNA short read mapping. M Alser, H Hassan, H Xin, O Ergin, O Mutlu‡, **C Alkan**‡. *Bioinformatics*, Nov 1; 33(21): 3335-63, 2017. *Preprint: arXiv:1604.01789.*

Toolkit for automated and rapid discovery of structural variants. A Soylev, C Kockan, F Hormozdiari‡, **C Alkan**‡. *Methods*, Oct 1, 129:3-7, 2017.

Discovery and genotyping of novel sequence insertions in many sequenced individuals. P Kavak, YY Lin, I Numanagić, H Asghari, T Güngör, **C Alkan**‡, F Hach‡. *Bioinformatics* (ISMB-ECCB 2017 issue), Jul 15; 33 (14): i161-i169, 2017.

presented at the 25th Annual International Conference on Intelligent Systems for Molecular Biology and 16th European Conference on Computational Biology (ISMB/ECCB 2017), July 21-25, 2017, Prague, Czech Republic.

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presented at the Ninth Annual International Conference on Research in Computational Molecular Biology (RECOMB 2005 - LNBI 3500), pp 152-171, May 14-18, 2005 Cambridge, MA, USA, 2005.

Manipulating multiple sequence alignments via MaM and WebMaM. **C Alkan**, E Tüzün, J Buard, F Lethiec, EE Eichler, JA Bailey, SC Şahinalp. *Nucleic Acids Research*, Jul 1; 33 (Web Server issue): W295-W298, 2005.

The role of unequal crossover in alpha-satellite DNA evolution: a computational analysis. **C Alkan**, EE Eichler, JA Bailey, SC Şahinalp, E Tüzün. *Journal of Computational Biology*, 11(5): 933-944, 2004.

presented at the 13th International Conference on Genome Informatics, December 16-18, 2002, Tokyo, Japan, 18: 93-102, 2002.

The structure and evolution of centromeric transition regions within the human genome. X She, JE Horvath, Z Jiang, G Liu, T S Furey, L Christ, R Clark, T Graves, CL Gulden, **C Alkan**, JA Bailey, SC Şahinalp, M Rocchi, D Haussler, RK Wilson, W Miller, S Schwartz, EE Eichler. *Nature*, Aug 19, 2004; 430(7002): 857-64, 2004.

Analysis of primate genomic variation reveals a repeat-driven expansion of the human genome. G Liu, NISC Comparative Sequencing Program, S Zhao, JA Bailey, SC Şahinalp, **C Alkan**, E Tüzün, ED Green, EE Eichler. *Genome Research*, 13(3): 358-368, 2003.

Divergent origins and concerted expansion of two segmental duplications on chromosome 16. EE Eichler, ME Johnson, **C Alkan**, E Tüzün, C Şahinalp, D Misceo, N Archidiacono, and M Rocchi. *Journal of Heredity*, 92: 462-468, 2001.

Book Chapters

Whole-genome shotgun sequence CNV detection using read depth. F Kahveci, **C Alkan**. In: Bickhart D (eds) *Copy Number Variants. Methods in Molecular Biology*, vol 1833. Humana Press, New York, NY, pp 61-72, 2018.

Peer-Reviewed Conference Proceedings

Note that selected papers accepted to some conferences, such as ISMB and RECOMB, are published in special issues of journals. Such publications are listed under Journal Publications with a note to the conference above. Joint first authors are marked with * in case of equal contribution. Joint last authors are marked with ‡.

KENT: fast pathogen screening with NVIDIA Jetson cluster in resource-constrained areas. A İçöz, R Roman-Brenes, Z Bingöl, **C Alkan**. 1st RECOMB Satellite Conference on Hardware Acceleration of Bioinformatics Workloads (RECOMB-Arch 2026), May 24-25, 2026, Thessaloniki, Greece.

In-depth data movement characterization of DNA read mapping algorithms. B Şahin, E İlgün, K Zambaku, ÖY Öztürk, **C Alkan**. 1st RECOMB Satellite Conference on Hardware Acceleration of Bioinformatics Workloads (RECOMB-Arch 2026), May 24-25, 2026, Thessaloniki, Greece.

SeGraM: a universal hardware accelerator for genomic sequence-to-graph and sequence-to-sequence mapping. D Senol Cali, K Kanellopoulos, J Lindegger, Z Bingöl, GS Kalsi, Z Zuo, C Firtina, MB Cavlak, J Kim, N Mansouri Ghiasi, G Singh, J Gómez-Luna, N Almadhoun Alserr, M Alser, S Subramoney, **C Alkan**, S Ghose, O Mutlu. *49th Annual International Symposium on Computer Architecture (ISCA 2022)*, pp 638-655, New York, NY, United States, June 18-22, 2022. *Preprint: arXiv:2205.05883.*

GenASM: a high-performance, low-power approximate string matching acceleration framework for genome sequence analysis. D Senol Cali, GS Kalsi, Z Bingöl, C Firtina, L Subramanian, JS Kim, R Ausavarungnirun, M Alser, J Gómez-Luna, A Boroumand, A Nori, A Scibisz, S Subramoney, **C Alkan**, S Ghose, O Mutlu. *53rd IEEE/ACM International Symposium on Microarchitecture (MICRO 2020)*, October 17-21, 2020, Athens, Greece. *Preprint: arXiv:2009.07692.*

Can you really anonymize the donors of genomic data in today's digital world? M Alser, N Almadhoun, A Nouri, **C Alkan**‡, and E Ayday‡. *10th International Workshop on Data Privacy Management (DPM 2015)*, September 21-22, 2015, Vienna, Austria.

Improving genome assemblies using multi-platform sequence data. P Kavak, B Ergüner, D Üstek, B Yüksel, MŞ. Sağıroğlu, T Güngör and **C Alkan**. *12th Computational Intelligence methods for Bioinformatics and Biostatistics (CIBB 2015)*, September 10-12, 2015, Naples, Italy.

RNA secondary structure prediction via energy density minimization. **C Alkan***, E Karakoç*, SC Şahinalp, P Unrau, H A Ebhardt, K Zhang, J Buhler. *Proc. of the Tenth Annual International Conference on Research in Computational Molecular Biology (RECOMB 2006 - LNBI 3909)*, pp. 130-142, April 2-5, 2006 Venice, Italy, 2006.

Unpublished Conference Talks and Posters

POACHA: partial order alignment computation hardware acceleration. Z Akcil, B Şahin, T Tanriverdi, Z Bingöl, **C Alkan**. Third Workshop on Architecture for Health (Arch4Health 2026) in conjunction with ACM International Conference on Supercomputing 2026 (ICS 2026), July 6, 2026, Belfast, Northern Ireland, UK. *Oral presentation by C Alkan.*

In-depth data movement characterization of DNA read mapping algorithms. B Şahin, E İlgün, K Zambaku, ÖY Öztürk, **C Alkan**. Third Workshop on Architecture for Health (Arch4Health 2026) in conjunction with ACM International Conference on Supercomputing 2026 (ICS 2026), July 6, 2026, Belfast, Northern Ireland, UK. *Oral presentation by B Şahin.*

FUSOR: Space-efficient taxonomic classification using hybrid hierarchical interleaved binary fuse filters with a stash. T Okçu, **C Alkan**. 30th Annual International Conference on Research in Computational Biology (RECOMB 2026), May 26-29, 2026, Thessaloniki, Greece. *Poster presentation by T Okçu.*

KENT: fast pathogen screening with NVIDIA Jetson cluster in resource-constrained areas. A İçöz, R Roman-Brenes, Z Bingöl, **C Alkan**. 30th Annual International Conference on Research in Computational Biology (RECOMB 2026), May 26-29, 2026, Thessaloniki, Greece. *Poster presentation by A İçöz.*

In-depth data movement characterization of DNA read mapping algorithms. B Şahin, E İlgün, K Zambaku, ÖY Öztürk, **C Alkan**. 30th Annual International Conference on Research in Computational Biology (RECOMB 2026), May 26-29, 2026, Thessaloniki, Greece. *Poster presentation by K Zambaku.*

FUSOR: Space-efficient taxonomic classification using hybrid hierarchical interleaved binary fuse filters with a stash. T Okçu, **C Alkan**. 16th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-Seq 2026), May 24-25, 2026, Thessaloniki, Greece. *Oral presentation by T Okçu.*

De Bruijn graphs for pangenomics: in-depth performance benchmarking of de Bruijn graph-based tools for read mapping. Z Bingöl, B Şahin, K Koliogeorgi, K Zambaku, R Roman-Brenes, C Fırtına, O Mutlu, **C Alkan**. 15th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-Seq 2025), April 24-25, 2025, Seoul, South Korea. *Oral presentation by Z Bingöl.*

Hardware/software co-design for sequence-to-pangenome mapping. Z Bingöl, C Fırtına, K Koliogeorgi, R Roman-Brenes, O Mutlu, **C Alkan**. 24th Genome Informatics Conference, November 13-15, 2024, Hinxton, England. *Poster presentation by Z Bingöl; Best Poster Award winner.*

SeGraM: a universal hardware accelerator for genomic sequence-to-graph and sequence-to-sequence mapping. D Senol Cali, K Kanellopoulos, J Lindegger, Z Bingöl, GS Kalsi, Z Zuo, C Firtina, MB Cavlak, JS Kim, N Mansouri Ghiasi, G Singh, J Gómez Luna, N Almadhoun Alserr, M Alser, S Subramoney, **C Alkan**, S Ghose, O Mutlu. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by D Senol Cali.*

Copy number estimation using Counting Bloom Filters in *de novo* assembled genomes. K Zambaku, R Roman-Brenes, ÖY Öztürk, **C Alkan**, I Birol. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by K Zambaku.*

Identification of protein-protein interaction hubs in multiple sclerosis. G Yazıcı, B Kurt Vatandaslar, E Karakoç, B Kerman, **C Alkan**. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by G Yazıcı.*

Whole genome alignment via Alternating Lyndon Tree Factorization. MS Aydın and **C Alkan**. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by MS Aydın.*

Characterization of alignment and search algorithms for short read, long read, and graph mappers. E İlgün, ÖY Öztürk, K Zambaku, J Gómez Luna, M Alser, R Roman-Brenes, **C Alkan** and The BioPIM Project. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by E İlgün.*

Pairwise sequence alignment with block and character edit operations. AC Alıcıoğlu, MS Aydın, **C Alkan**. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by MS Aydın.*

AirLift: a fast and comprehensive technique for remapping alignments between reference genomes. JS Kim, C Firtina, MB Cavlak, D Senol Cali, N Hajinazar, M Alser, **C Alkan**, O Mutlu. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by C Firtina.*

Fast identification of sepsis antibiotic resistance. R Roman-Brenes, EG Güliter, **C Alkan**. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by R Roman-Brenes.*

TargetCall: eliminating the wasted computation in basecalling via pre-basecalling filtering. MB Cavlak, G Singh, M Alser, C Firtina, J Lindegger, M Sadrosadati, N Mansouri Ghiasi, **C Alkan**, O Mutlu. 27th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2023), April 16-19, 2023, İstanbul, Türkiye. *Poster presentation by MB Cavlak.*

Pairwise sequence alignment with block and character edit operations. AC Alıcıoğlu, MS Aydın, **C Alkan**. 13th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-Seq 2023), April 14-15, 2023, İstanbul, Türkiye. *Oral presentation by MS Aydın.*

SeGraM: a universal hardware accelerator for genomic sequence-to-graph and sequence-to-sequence mapping. D Senol Cali, K Kanellopoulos, J Lindegger, Z Bingöl, GS Kalsi, Z Zuo, C Firtina, MB Cavlak, JS Kim, N Mansouri Ghiasi, G Singh, J Gómez Luna, N Almadhoun Alserr, M Alser, S Subramoney, **C Alkan**, S Ghose, O Mutlu. 13th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-Seq 2023), April 14-15, 2023, İstanbul, Türkiye. *Poster presentation by D Senol Cali.*

Using minimizer interarrival distances for read-until human read detection from blood samples sequenced by Oxford Nanopore. S Barazandeh, MS Aydin, B Ucar, **C Alkan**, I Birol. 13th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-Seq 2023), April 14-15, İstanbul, 2023, Türkiye. *Poster presentation by MS Aydın.*

Characterization of alignment and search algorithms for short read, long read, and graph mappers. E İlgün, ÖY Öztürk, K Zambaku, J Gómez Luna, M Alser, R Roman-Brenes, **C Alkan**n and The BioPIM Project. 13th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-Seq 2023), April 14-15, 2023, İstanbul, Türkiye. *Poster presentation by E İlgün.*

Copy number estimation using Counting Bloom Filters in *de novo* assembled genomes. K Zambaku, R Roman-Brenes, ÖY Öztürk, **C Alkan**, I Birol. 13th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-Seq 2023), April 14-15, 2023, İstanbul, Türkiye. *Poster presentation by K Zambaku.*

Identification of protein-protein interaction hubs in multiple sclerosis. G Yazıcı, B Kurt Vatandaslar, E Karakoç, B Kerman, **C Alkan**. 21st European Conference on Computational Biology (ECCB 2022), September 12-21, 2022, Sitges, Spain. *Poster presentation by G Yazıcı.*

Polishing copy number variant calls on exome sequencing data via deep learning. F Özden, **C Alkan**[‡], AE Çiçek[‡]. 29th Annual International Conference Intelligent Systems for Molecular Biology (ISMB/ECCB 2021), *HiTSeq Track*, July 27, 2021, held virtually. *Oral presentation by F Özden.*

GateKeeper-GPU: fast and accurate pre-alignment filtering in short read mapping. Z Bingol, M Alser, O Ozturk, **C Alkan**. 20th IEEE International Workshop on High Performance Computational Biology (HiCOMB), May 17, 2021, Portland, OR, United States. *Oral presentation by Z Bingol.*

Discovery of structural variations in ancient genomes. A Soylev, **C Alkan**, M Somel. Twelfth International Symposium on Health Informatics and Bioinformatics (HIBIT), October 17-18, 2019, İzmir, Turkey. *Oral presentation by A Soylev.*

Characterization of large-scale structural variants using Linked-Reads. F Karaoglanoglu, C Ricketts, E Ebren, M Eslami Rasekh, I Hajirasouliha, **C Alkan**. 27th Annual International Conference Intelligent Systems for Molecular Biology (ISMB/ECCB 2019), *HiTSeq Track*, July 23, 2019, Basel, Switzerland. *Oral presentation by C Alkan.*

SneakySnake: a fast and efficient pre-alignment filter for accelerating approximate string matching. M Alser, **C Alkan**, O Mutlu. 27th Annual International Conference Intelligent Systems for Molecular Biology (ISMB/ECCB 2019), *HiTSeq Track*, July 22-23, 2019, Basel, Switzerland. *Poster presentation by M Alser.*

BitMAC: an in-memory accelerator for bit vector-based sequence alignment of both short and long genomic reads. D Senol Cali, C Firtina, JS Kim, Z Bingol, M Alser, **C Alkan**, S Ghose, O Mutlu. 27th Annual International Conference Intelligent Systems for Molecular Biology (ISMB/ECCB 2019), *HiTSeq Track*, July 22-23, 2019, Basel, Switzerland. *Poster presentation by C Firtina.*

Apollo: a sequencing-technology-independent, scalable, and accurate assembly polishing algorithm. C Firtina, JS Kim, M Alser, D Senol Cali, AE Cicek, **C Alkan**, O Mutlu. 27th Annual International Conference Intelligent Systems for Molecular Biology (ISMB/ECCB 2019), *HiTSeq Track*, July 22-23, 2019, Basel, Switzerland. *Poster presentation by C Firtina.*

Graph based plasmid identification in bacteria. F Kahveci, Ö. Kalay, A Jain, TF Jesus, JA Carriço, **C Alkan**. Eleventh International Symposium on Health Informatics and Bioinformatics (HIBIT), October 25-27, 2018, Antalya, Turkey. *Oral presentation by F Kahveci.*

Hercules: a profile HMM-based hybrid error correction algorithm for long reads. C Firtina, Z Bar-Joseph, **C Alkan**, AE Cicek. 26th Annual International Conference on Intelligent Systems for Molecular Biology (ISMB 2018), July 7-10, 2018, Chicago, IL, United States. *Oral presentation by AE Cicek.*

Nanopore sequencing technology and tools for genome assembly: computational analysis of the current state, bottlenecks, and future directions. D Senol Cali, JS Kim, S Ghose, **C Alkan**, O Mutlu. 22nd Annual International Conference on Research in Computational Molecular Biology (RECOMB 2018), April 21-24, 2018, Paris, France. *Poster presentation by D Senol Cali.*

Accelerating approximate pattern matching with processing-in-memory (PIM) and single-instruction multiple-data (SIMD) programming. D Senol Cali, Z Bingol, JS Kim, R Ausavarungnirun, S Ghose, **C Alkan**, O Mutlu. The Eighth RECOMB Satellite Workshop on Massively Parallel Sequencing, April 19-20, 2018, Paris, France. *Poster presentation by D Senol Cali. Best Poster Award Winner.*

Algorithms for structural variation discovery using hybrid sequencing technologies. E Ebren, AB Dinçer, **C Alkan**. 25th Annual International Conference on Intelligent Systems for Molecular Biology and 16th European Conference on Computational Biology (ISMB/ECCB 2017), July 21-25, 2017, Prague, Czech Republic. *Poster presentation by E Ebren.*

Discovery of large genomic inversions using long-range information. F Karaoğlanoğlu, M Eslami Rasekh, **C Alkan**. Tenth International Symposium on Health Informatics and Bioinformatics (HIBIT), June 28-30, 2017, Güzelyurt, Turkish Republic of North Cyprus. *Oral presentation by F Karaoğlanoğlu.*

A profile HMM-based hybrid error correction algorithm for long sequencing reads. C Fırtına, Z Bar-Joseph, AE Çiçek, **C Alkan**. Tenth International Symposium on Health Informatics and Bioinformatics (HIBIT), June 28-30, 2017, Güzelyurt, Turkish Republic of North Cyprus. *Poster presentation by C Fırtına.*

LEAP: a generalization of the Landau-Vishkin algorithm with custom gap penalties. H Xin, JS Kim, S Nahar, C Kingsford, **C Alkan**, O Mutlu. The Seventh RECOMB Satellite Workshop on Massively Parallel Sequencing, May 7-8, 2017, Hong Kong. *Oral presentation by H Xin.*

Yapısal varyasyonların karakterizasyonu. A Söylev, **C Alkan**. 12th National Medical Genetics Congress, October 7, 2016, Çeşme, Turkey. *Oral presentation by A Söylev (in Turkish).*

A new inference attack against kin genomic privacy. F Balci, H Kulan, **C Alkan**‡, and E Ayday‡. Privacy-aware computational genomics (PRIVAGEN 2015), September 8, 2015, Tokyo, Japan. *Oral and poster presentation by F Balcı.*

Identifying anonymous donors of genetic information. M Alser, N Almadhoun, A Nouri, **C Alkan**‡, and E Ayday‡. Privacy-aware computational genomics (PRIVAGEN 2015), September 8, 2015, Tokyo, Japan. *Poster presentation by M Alser.*

BioPeer: a fast and secure peer-to-peer data sharing tool. C Oge, FT Dogan, G Goktepe, F Koc, C Sevim, **C Alkan**. The 6th Conference on High Throughput Sequencing Analysis and Algorithms (HiTSeq 2015), July 10-11, 2015, Dublin, Ireland. *Poster presentation by C Öge.*

Coinami: a cryptocurrency with DNA sequence alignment as proof-of-work. AM Ileri, HI Ozercan, A Gundogdu, AK Senol, MY Ozkaya, **C Alkan**. The 6th Conference on High Throughput Sequencing Analysis and Algorithms (HiTSeq 2015), July 10-11, 2015, Dublin, Ireland. *Oral and poster presentation by C Alkan.*

Optimal Seed Solver: optimizing seed selection in read mapping. H Xin, S Nahar, R Zhu, J Emmons, G Pekhimenko, C Kingsford, **C Alkan**‡, O Mutlu‡. The 6th Conference on High Throughput Sequencing Analysis and Algorithms (HiTSeq 2015), July 10-11, 2015, Dublin, Ireland. *Poster presentation by H Xin.*

Discovery of large genomic inversions using pooled clone sequencing. ME Rasekh, G Chiatante, M Miroballo, J Tang, M Ventura, CT Amemiya, EE Eichler, F Antonacci‡, **C Alkan**‡. The 6th Conference on High Throughput Sequencing Analysis and Algorithms (HiTSeq 2015), July 10-11, 2015, Dublin, Ireland. *Poster presentation by M Eslami Rasekh. Preprint: bioRxiv 015156.*

A hypergraph-based model for hybrid de novo assembly. S Ashyralyyev, C Firtina, C Aykanat, **C Alkan**. Bertinoro Computational Biology Meeting, June 17, 2015, Bertinoro, Italy. *Oral presentation by S Ashyralyyev.*

Massively parallel mapping of next generation sequence reads using GPUs. A Nouri, RO Selvitopi, O Ozturk, O Mutlu‡, **C Alkan**‡. The 20th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2015), March 14-18, 2015, İstanbul, Turkey. *Short talk and poster presented by A Nouri.*

Preprints

Preprints later published in a journal or a peer-reviewed conference are moved from this section to their respective sections.

Toward space-optimal approximate colored indexes at scale. T Okcu, **C Alkan**, A Limasset. *bioRxiv 2026.11.05.764633*, posted November 6, 2026. doi: 10.64898/2026.11.05.764633.

Somatic structural variation discovery on cancer pangenome graphs. M Kara, A Ashyralyyev, **C Alkan**. *bioRxiv 2026.11.05.764632*, posted November 6, 2026. doi: 10.64898/2026.11.05.764632.

In-depth data movement characterization of DNA read mapping algorithms. B Şahin, E İlgün, K Zambaku, **C Alkan**. *arXiv:2611.03400*, posted November 2, 2026. doi: 10.48550/arXiv.2611.03400.

KENT: fast pathogen screening with NVIDIA Jetson cluster in resource-constrained areas. A Icoz, R Roman-Brenes, Z Bingöl, **C Alkan**. *bioRxiv 2026.09.28.754999*, posted October 2, 2026. doi: 10.64898/2026.09.28.754999.

FASTR: reimagining FASTQ via compact image-inspired representation. A Tkachenko, S Salem, AE Adeniyi, Z Bingöl, MN Uddin, A Prasanna, A Zelikovsky, S Mangul, **C Alkan**, M Alser. *bioRxiv 2026.01.22.701172*, posted January 23, 2026. doi: 10.64898/2026.01.22.701172.

GenCore: Genomic distance estimation using Locally Consistent Parsing. A Ashyralyyev, E Sirvan, E Ilgun, S Malikic, T Batu, SC Sahinalp, **C Alkan**. *bioRxiv 2026.01.10.698768*, posted January 12, 2026. doi: 10.64898/2026.01.10.698768.

Robust software development practices improve citations of RNA-seq tools. S Sharma, E İlgün, T Okçu, V Ostash, V Bashynska, **C Alkan**, E Myoung, NN Alpana, K Chhugani, D Deshpande, AAT Bui, K Ellrott, O Burduniuc, TV Tatarinova, V Munteanu, PP Łabaj, D Ciorba, V Bostan, M Dimian, M Pellegrini, M Alser, S Mangul. *bioRxiv 2025.09.05.674580*, posted September 10, 2025. doi: 10.1101/2025.09.05.674580.

Pairwise sequence alignment with block and character edit operations. AC Alıcıoğlu, **C Alkan**. *arXiv:2311.11082*, posted Nov 18, 2023. doi: 10.48550/arXiv.2311.11082

GateSeeder: Near-memory CPU-FPGA acceleration of short and long read mapping. J Eudine, M Alser, G Singh, **C Alkan**, O Mutlu. *arXiv:2309.17063*, posted Sep 29, 2023. doi: 10.48550/arXiv.2309.17063.

LEAP: a generalization of the Landau-Vishkin algorithm with custom gap penalties. H Xin, J Kim, S Nahar, **C Alkan**[‡], O Mutlu[‡]. *bioRxiv 133157*, posted May 7, 2017. doi: 10.1101/133157.

Coinami: a cryptocurrency with DNA sequence alignment as proof-of-work. AM Ileri, HI Ozercan, A Gundogdu, AK Senol, MY Ozkaya, **C Alkan**. *arXiv:1602.03031*, posted Feb 9, 2016. doi: 10.48550/arXiv.1602.03031.

Software

- mrCaNaVaR: segmental duplication and absolute copy number prediction from read depth.
  `https://github.com/BilkentCompGen/mrcanavar`
- SCALCE: tool to compress Illumina FASTQ files.
  `https://github.com/sfu-compbio/scalce`
- VALOR: structural variation discovery using long-range information.
  `https://github.com/BilkentCompGen/valor`
- TARDIS: Toolkit for the automated and rapid discovery of structural variants.
  `https://github.com/BilkentCompGen/tardis`
- Shouji: fast and efficient computation of banded sequence alignment.
  `https://github.com/CMU-SAFARI/Shouji`
- SneakySnake: approximate alignment computation on CPU, GPU, and FPGA by reducing the alignment problem to a single net routing problem.
  `https://github.com/CMU-SAFARI/SneakySnake`
- BISER: Characterization of segmental duplication evolutionary structure across genome assemblies.
  `https://github.com/0xTCG/biser`
- RinsLR: Discovery of mid-range novel sequence insertions using long-read sequencing.
  `https://github.com/vpc-ccg/rinslr`
- DeCoNT: CNV prediction polisher on whole-exome sequencing data.
  `https://github.com/ciceklab/DECoNT`
- ECOLE: CNV prediction tool using whole-exome sequencing data.
  `https://github.com/ciceklab/ECOLE`

- GenASM: approximate string matching (ASM) acceleration framework for genome sequence analysis. `https://github.com/CMU-SAFARI/GenASM`
- SeGraM: universal genomic mapping accelerator that supports both sequence-to-graph mapping and sequence-to-sequence mapping for both short and long reads. `https://github.com/CMU-SAFARI/SeGraM`
- FastRemap: Tool for quickly remapping reads between genome assemblies. `https://github.com/CMU-SAFARI/FastRemap`
- BLEND: Fast, memory-efficient, and accurate mechanism to find fuzzy seed matches in genome analysis. `https://github.com/CMU-SAFARI/BLEND`
- AirLift: Fast and comprehensive technique for translating alignments between reference genomes. `https://github.com/CMU-SAFARI/AirLift`
- BriFin: Identification of bridges in intercellular PPI networks. `https://github.com/BilkentCompGen/brifin`
- TargetCall: Pre-basecalling filter for the Oxford Nanopore platform. `https://github.com/CMU-SAFARI/TargetCall`
- SABER: Sequence alignment using block edits and rearrangements. `https://github.com/BilkentCompGen/saber`
- LCPtools: Implementation of the Locally Consistent Parsing (LCP) algorithm for string processing. `https://github.com/BilkentCompGen/lcptools`
- Corallel: Fast alignment-free genomic distance estimation using LCP. `https://github.com/BilkentCompGen/corallel`
- LCPan: Variation Graph Construction Using Locally Consistent Parsing. `https://github.com/BilkentCompGen/lcpan`
- FAVeS: A lightweight SNP caller. `https://github.com/BilkentCompGen/faves`
- KoNuSeg: A fast, memory-efficient repeat detection using k-mer counting and Bloom filters. `https://github.com/BilkentCompGen/KoNuSeg`

Software (Deprecated / Not Maintained)

- Linden: Alternating Lyndon Factorization Tree for whole-genome alignments. `https://github.com/BilkentCompGen/linden`
- Pamir: novel sequence insertion discovery in multiple samples. `https://github.com/vpc-ccg/pamir`
- Hercules: Profile HMM-based hybrid error correction algorithm for long reads. `https://github.com/BilkentCompGen/hercules`
- Apollo: Profile HMM-based genome assembly polishing tool. `https://github.com/CMU-SAFARI/Apollo`
- GateKeeper: FPGA design for accelerating pre-alignment in DNA short read mapping. `https://github.com/BilkentCompGen/GateKeeper`
- GateKeeper-GPU: GPGPU design and implementation for the GateKeeper algorithm. `https://github.com/BilkentCompGen/GateKeeper-GPU`
- MAGNET: FPGA-based filtering strategy with high accuracy across different edit distance thresholds. `https://github.com/BilkentCompGen/MAGNET`
- taveRNA: RNA suite for RNA folding, RNA-RNA interaction prediction and search. `http://compbio.cs.sfu.ca/taverna`
- VariationHunter/CommonLAW: Structural variation calling algorithm for paired-end, next-generation sequencing data (replaced by TARDIS). `http://variationhunter.sourceforge.net`
- NovelSeq: a computational pipeline to detect novel sequence insertions using second-generation sequencing (replaced by Pamir). `http://novelseq.sourceforge.net`
- RepeatNet: an algorithm to characterize centromeric satellite sequences from paired-end sequence data. `https://github.com/BilkentCompGen/repeatnet`
- SEDEF: Characterization of segmental duplications within genome assemblies (replaced by BISER). `https://github.com/vpc-ccg/sedef/`
- Multiple alignment Manipulator (MaM). `https://github.com/BilkentCompGen/mam`

- micro-read Fast Alignment Search Tools (mrFAST, mrsFAST, drFAST, and sirFAST). `https://github.com/BilkentCompGen/mrfast`, `https://github.com/sfu-compbio/mrsfast`, `https://github.com/BilkentCompGen/drfast`, `https://github.com/BilkentCompGen/sirfast`